

Thm. If f is a continuous complex-valued function on $[a, b]$, then there exists a sequence of polynomials P_n such that

$$P_n \rightarrow f \text{ uniformly on } [a, b].$$

Proof. WLOG, we assume $[a, b] = [0, 1]$, and $f(0) = f(1) = 0$.

(This is possible: $g(x) = f(x) - f(0) - x(f(1) - f(0))$)

And, $f(x) = 0$ if $x \notin [0, 1]$.

Put $Q_n(x) = C_n \cdot (1-x^2)^n \quad (n=1, 2, 3, \dots)$

where C_n is chosen such that $\int_{-1}^1 Q_n(x) dx = 1$.

To magnitade C_n ,

$$\begin{aligned} \int_{-1}^1 (1-x^2)^n dx &= 2 \cdot \int_0^1 (1-x^2)^n dx \geq 2 \cdot \int_0^{\frac{1}{\sqrt{n}}} (1-x^2)^n dx \\ &\geq 2 \cdot \int_0^{\frac{1}{\sqrt{n}}} (1-nx^2) dx = 2nx \cdot \left[\frac{(1-x^2)^{n+1}}{-(n+1)} + 1 \right] \geq 0 \\ &= 2 \cdot \left(\frac{1}{\sqrt{n}} - \frac{n}{3} \cdot \frac{1}{(\sqrt{n})^3} \right) = \frac{4}{3} \frac{1}{\sqrt{n}} > \frac{1}{\sqrt{n}} \end{aligned}$$

so, $C_n < \sqrt{n}$. Now, for any $\delta > 0$,

$$Q_n(x) \leq \sqrt{n} \cdot (1-\delta^2)^n \quad (\delta \leq |x| \leq 1), \text{ so that } Q_n \rightarrow 0 \text{ uniformly in } \delta \leq |x| \leq 1$$

Now, set $P_n(x) = \int_{-1}^1 f(x+t) \cdot Q_n(t) dt \quad (0 \leq x \leq 1)$

Since $f(s) = 0$ if $s \notin [0, 1]$,

$$= \int_{-x}^{1-x} f(x+t) Q_n(t) dt$$

$$= \int_0^1 f(s) Q_n(s-x) ds \quad (x+t=s \Rightarrow dt=ds)$$

Given $\epsilon > 0$, choose $\delta > 0$ s.t. $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\epsilon}{2}$, since f is continuous on compact

$$|P_n(x) - f(x)| = \left| \int_0^1 [f(x+t) - f(x)] \cdot Q_n(t) dt \right| \leq \int_0^1 |f(x+t) - f(x)| \cdot Q_n(t) dt$$

$$\leq 2M \cdot \left(\int_{-1}^{-\delta} Q_n(t) dt + \int_{\delta}^1 Q_n(t) dt \right) + \frac{\epsilon}{2} \cdot \int_{-\delta}^{\delta} Q_n(t) dt \quad (M = \sup |f|)$$

$$\leq 4M \cdot \sqrt{n} (1-\delta^2)^n + \frac{\epsilon}{2} \rightarrow 0$$